

THORACIC WALL

Department of Anatomy

Thorax (1)

Third part

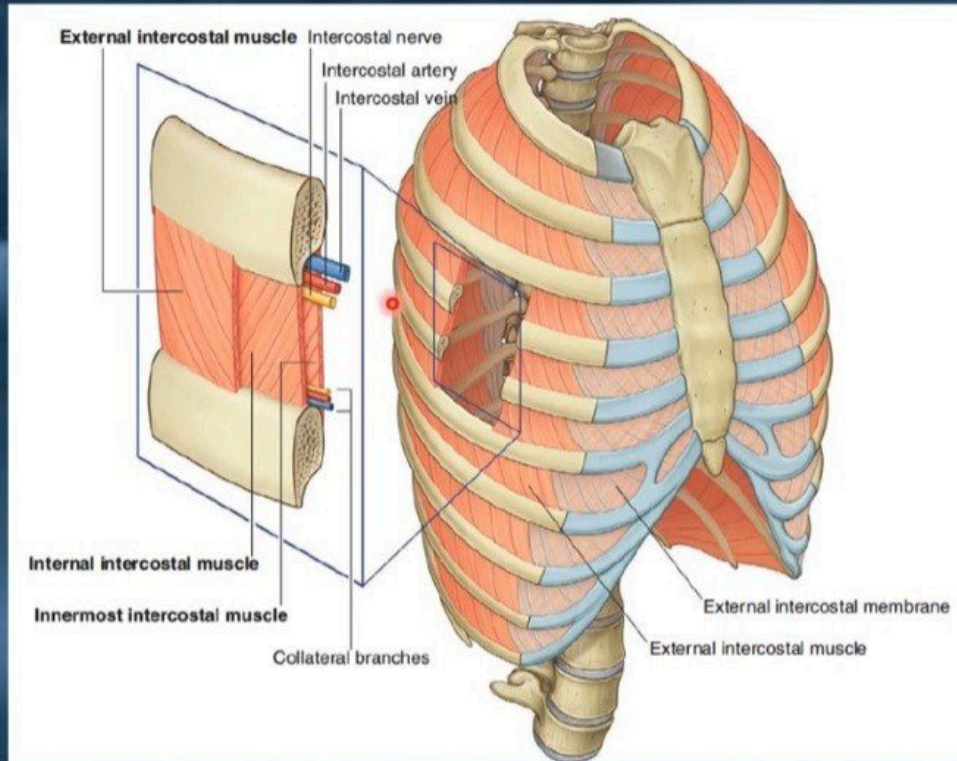
Dr. Osman Ahmed Osman

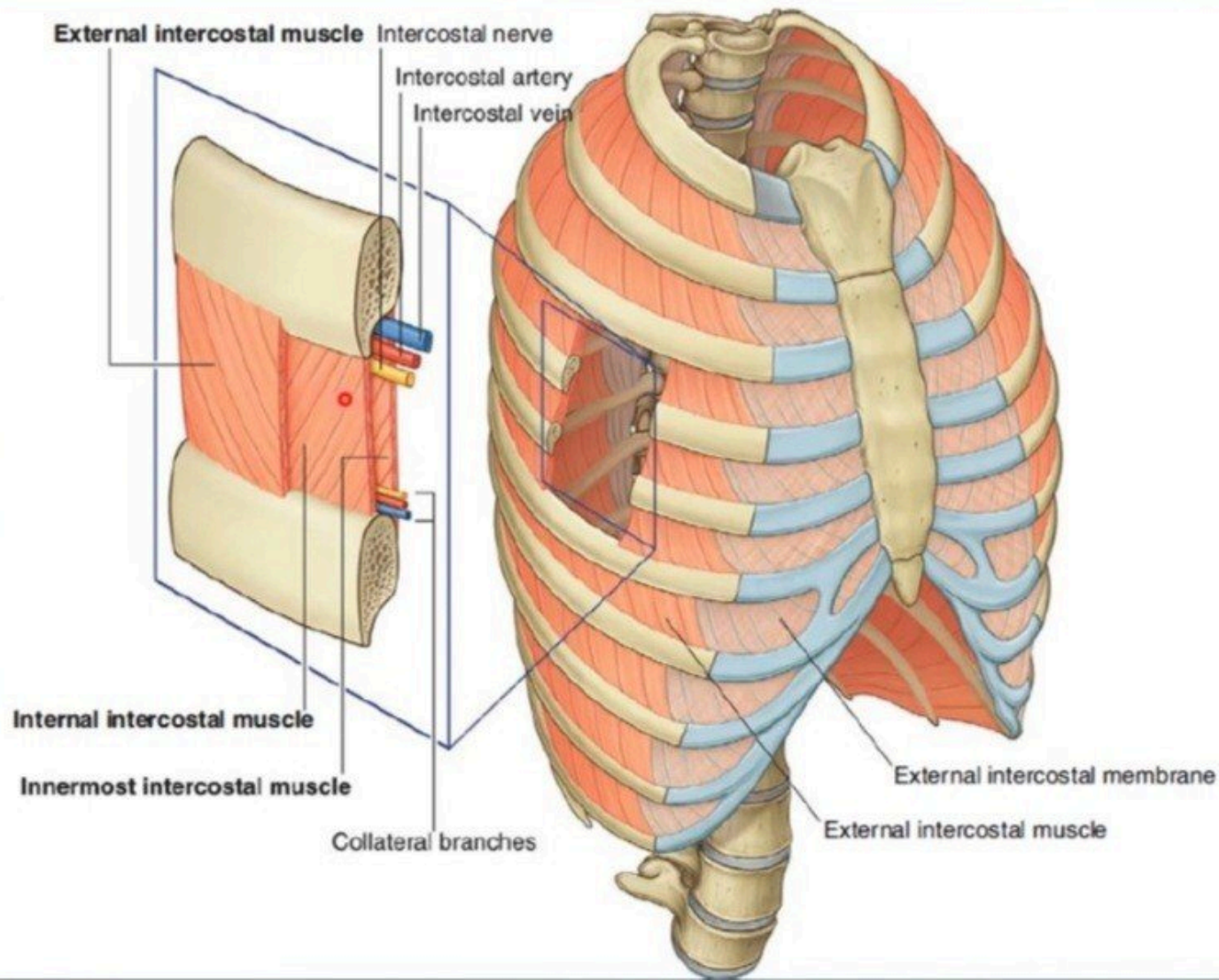
Lecture Points

- **Anatomy of the Thoracic wall and Thoracic Cavity.**
- **The superior thoracic aperture consists.**
- **The suprapleural membrane**
- **The bone Structure of the Thoracic Cage.**
 - **Sternum.**
 - **The sternal angle.**
 - **Ribs (Typical & Atypical).**
 - **Thoracic Vertebra (Typical & Atypical).**
 - **Joints (articulations) of the thorax**
- **Inferior Boundary (outlet) of the thorax.**
- **The diaphragm.**
 - ✓ **Physiological action of diaphragm.**
 - ✓ **Three Large Openings of Diaphragm.**
 - ✓ **arterial supply of the Diaphragm.**
- **The muscles of the thorax wall**
- **The arterial supply to the thoracic wall**
- **The venous drainage of the thoracic wall.**
- **lymphatic drainage of the thorax wall.**
- **Intercostal nerve (Typical & Atypical).**
- **Segmental innervation (dermatomes) of thoracic wall.**
- **herpes zoster the Spinal Ganglia**
- **Vertebral levels of sternum and transverse thoracic plane.**
- **Surface Anatomy of the Thoracic Wall Skeleton.**

The muscles of the thorax wall

- The external intercostal muscles.
- The internal intercostal muscles.
- The innermost intercostal muscles.
- The subcostal muscles.
- The transversus thoracis muscles.

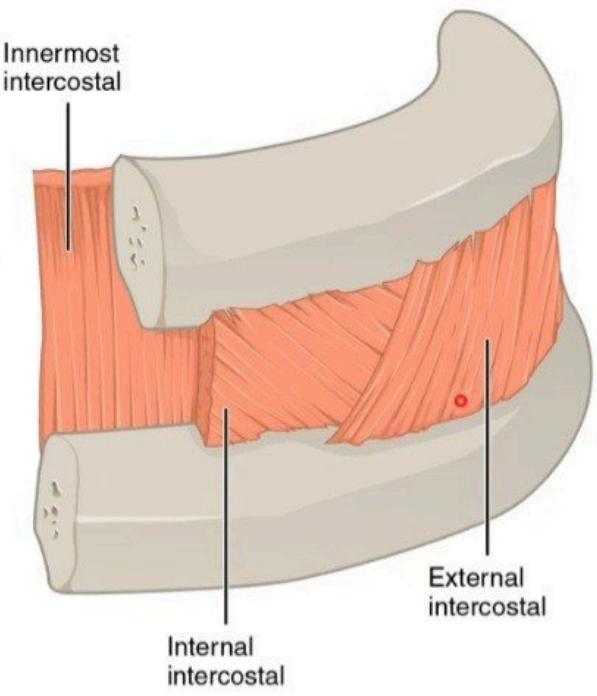
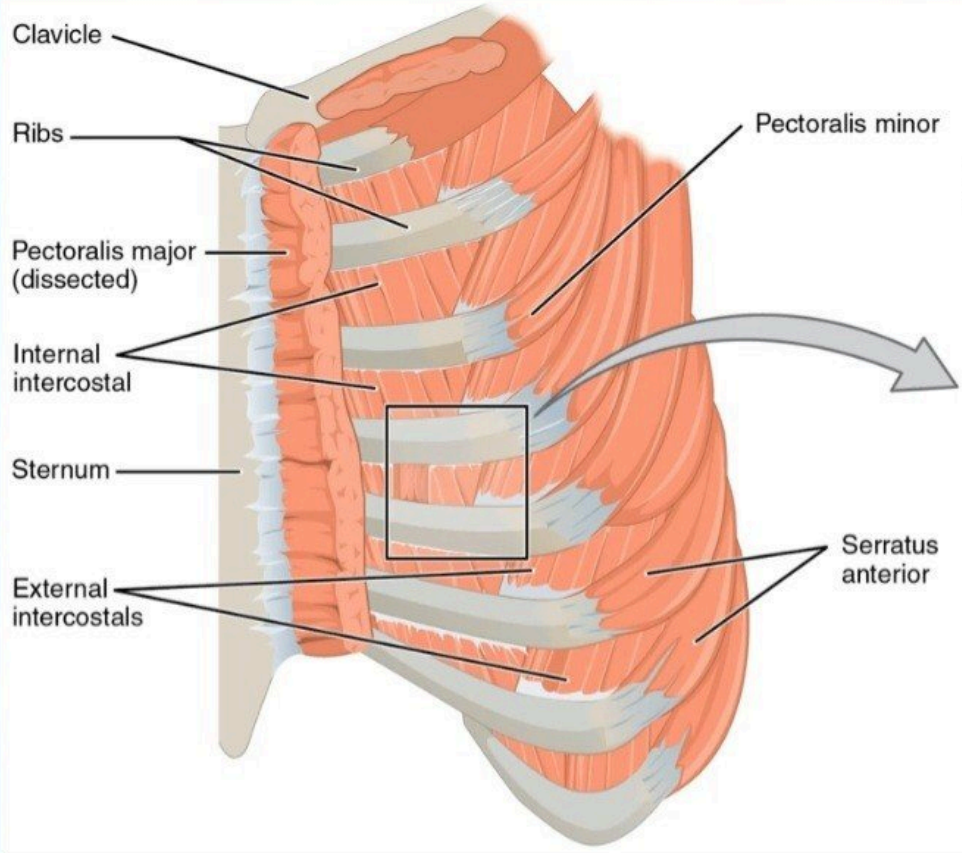


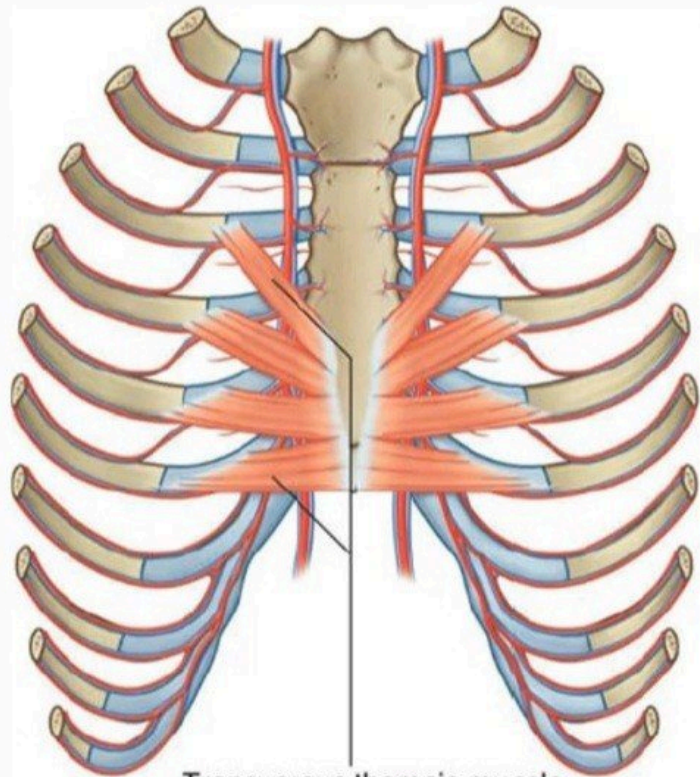


Muscles of the thoracic wall

Muscles of the thoracic wall

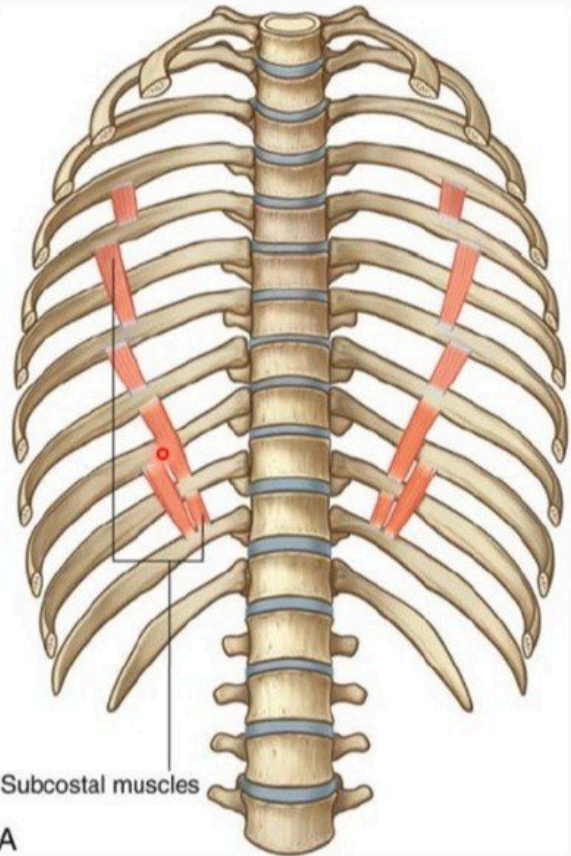
Muscle	Superior attachment	Inferior attachment	Innervation	Function
External intercostal	Inferior margin of rib above	Superior margin of rib below	Intercostal nerves; T1–T11	Most active during inspiration; supports intercostal space; moves ribs superiorly
Internal intercostal	Lateral edge of costal groove of rib above	Superior margin of rib below deep to the attachment of the related external intercostal	Intercostal nerves; T1–T11	Most active during expiration; supports intercostal space; moves ribs inferiorly
Innermost intercostal	Medial edge of costal groove of rib above	Internal aspect of superior margin of rib below	Intercostal nerves; T1–T11	Acts with internal intercostal muscles
Subcostales	Internal surface (near angle) of lower ribs	Internal surface of second or third rib below	Related intercostal nerves	May depress ribs
Transversus thoracis	Inferior margins and internal surfaces of costal cartilages of second to sixth ribs	Inferior aspect of deep surface of body of sternum, xiphoid process, and costal cartilages of ribs IV–VII	Related intercostal nerves	Depresses costal cartilages





Transversus thoracis muscle

B

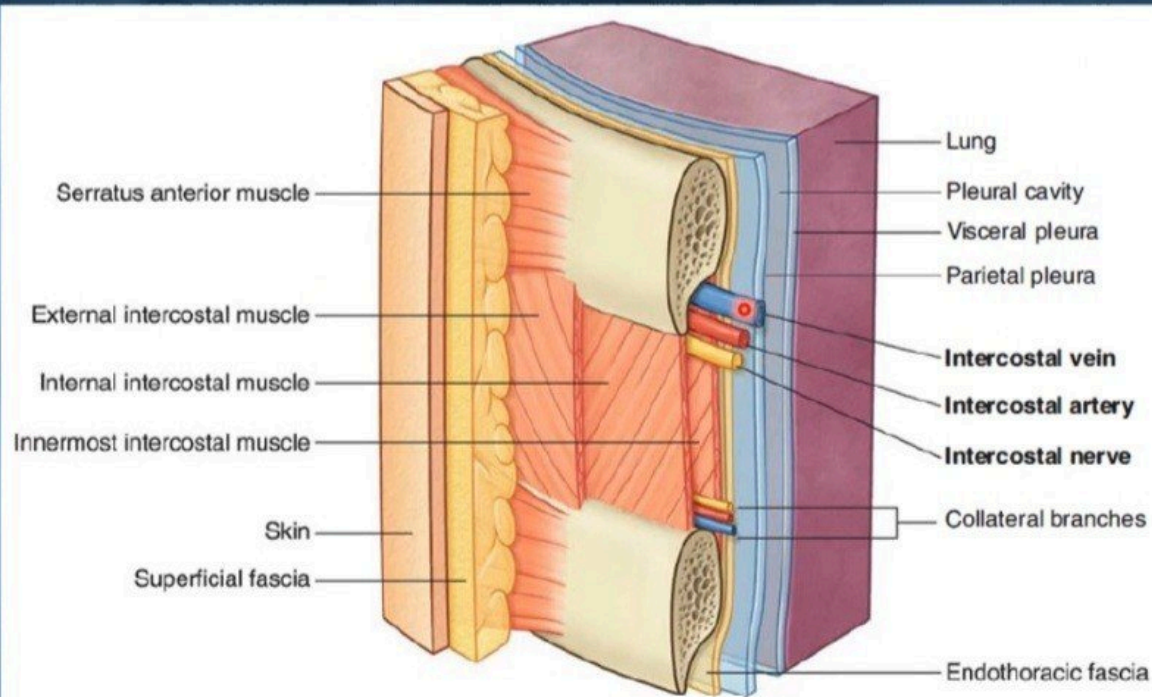


Subcostal muscles

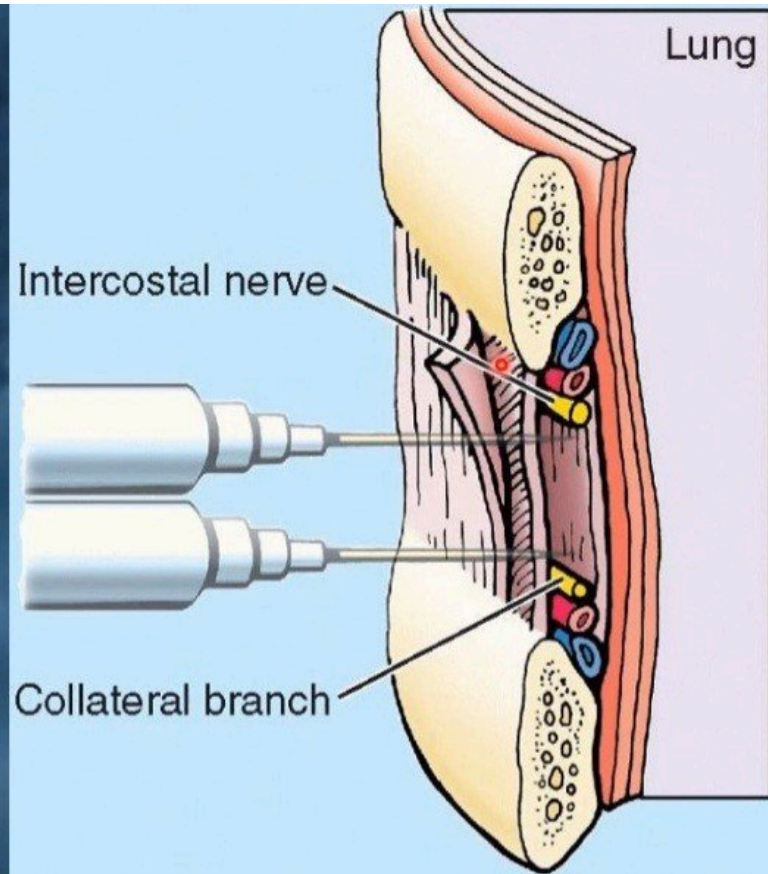
A

Neurovascular Bundle of Intercostals

- **VAN (vein, artery, nerve)**
 - Intercostal vein
 - Intercostal artery
 - Intercostal nerve
- **Sit in Subcostal Groove**
- Between **Internal Intercostal** and **Innermost intercostal muscles**



Site of injection



Posterior ramus of spinal nerve

Posterior intercostal artery and vein

Lateral branches of
intercostal nerve
and vessels

Intercostal nerve

Aorta

Costal groove

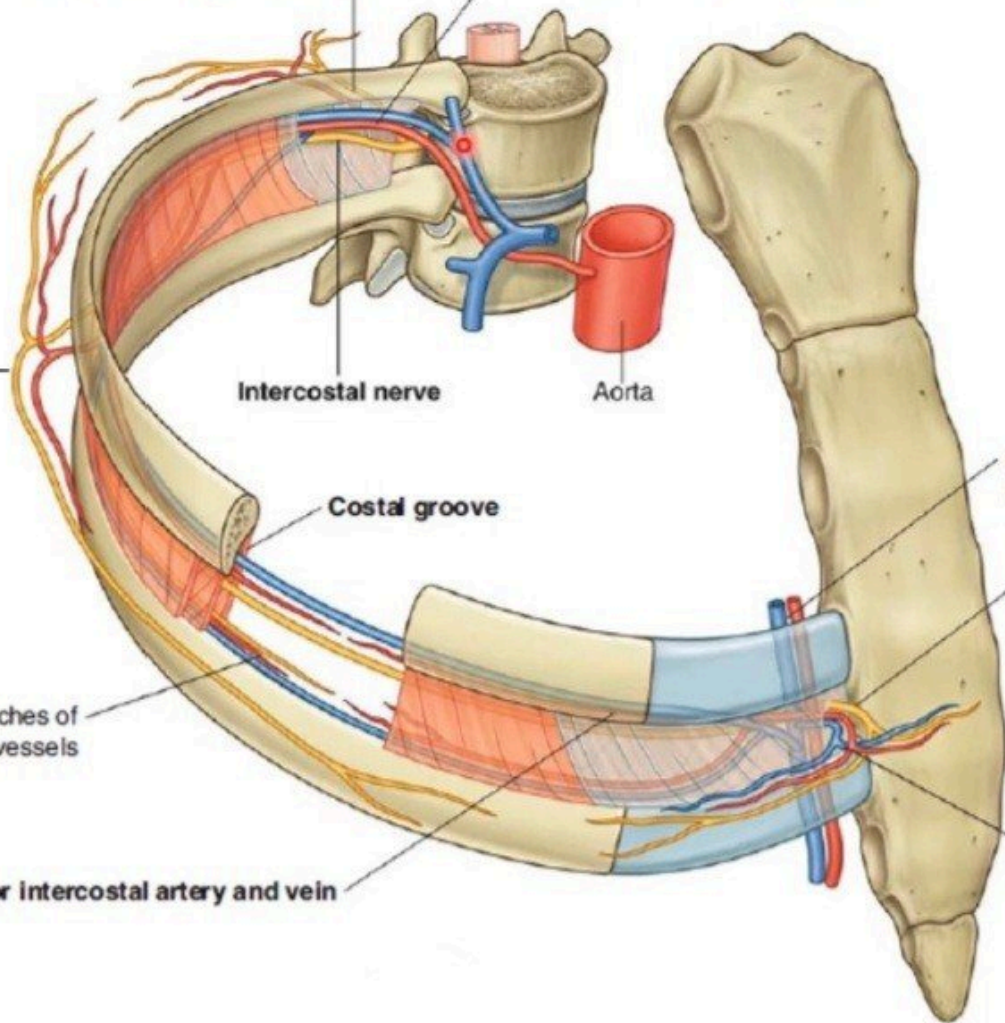
Internal thoracic artery and vein

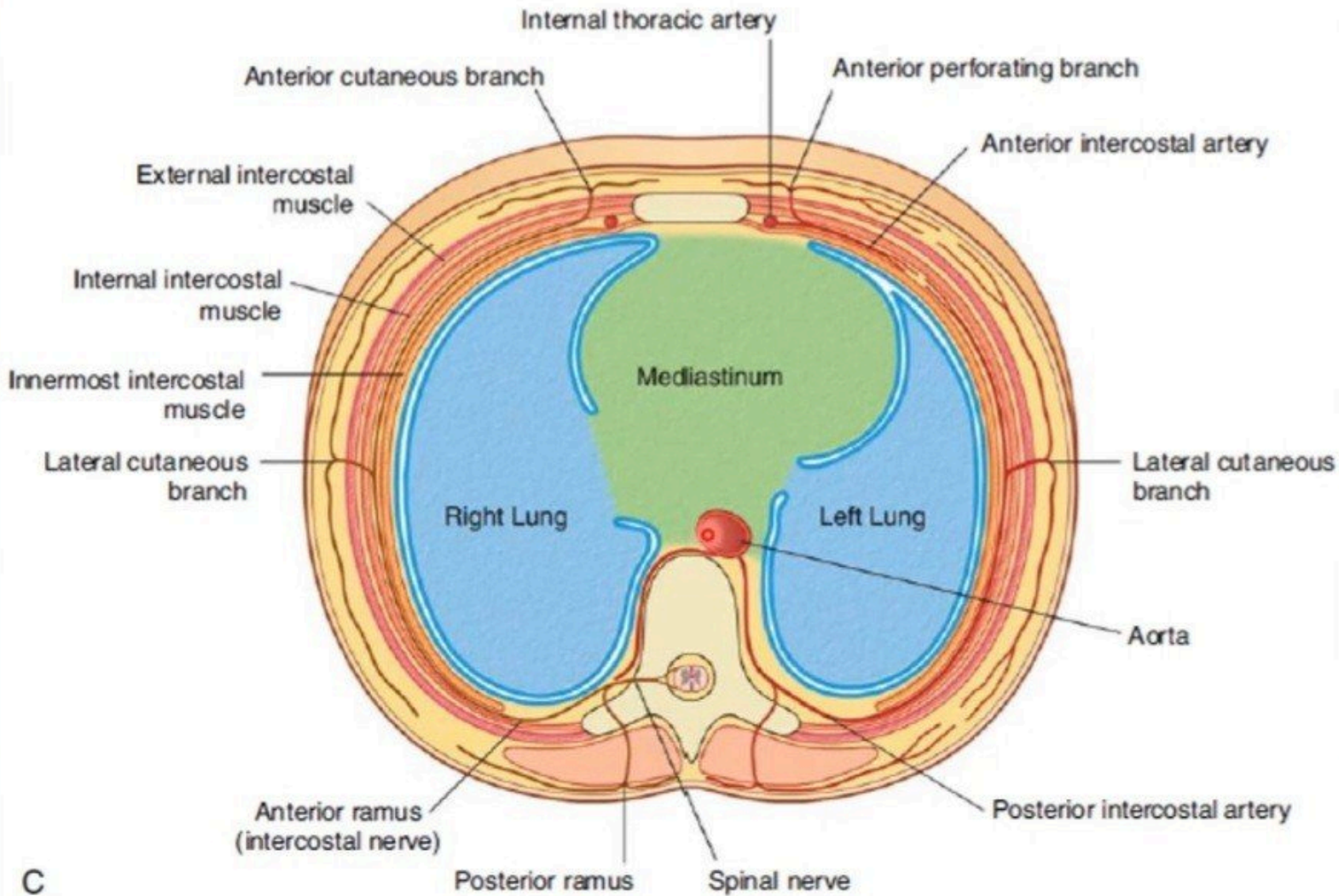
Anterior cutaneous
branch of
intercostal nerve

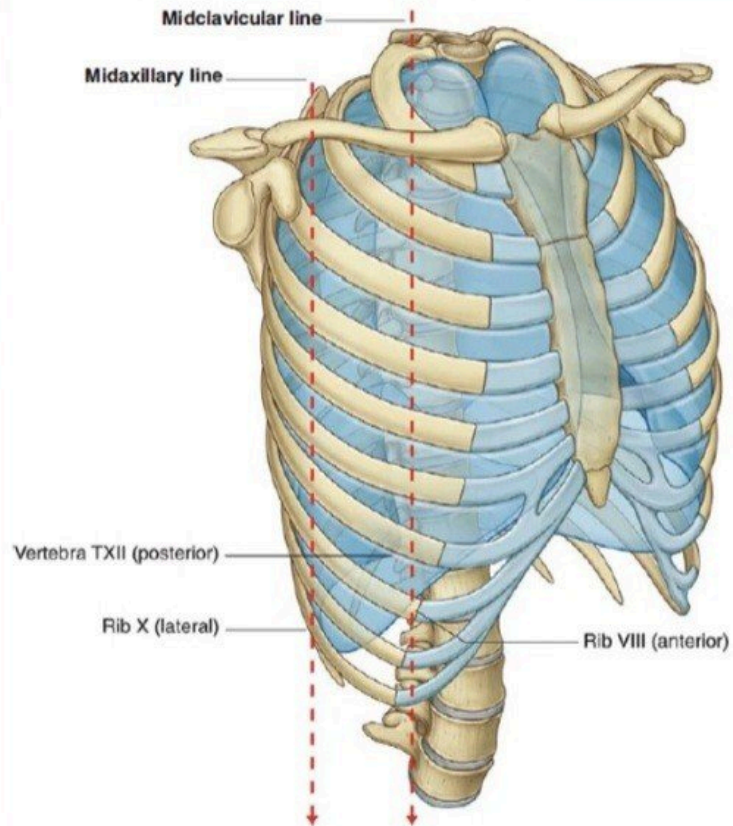
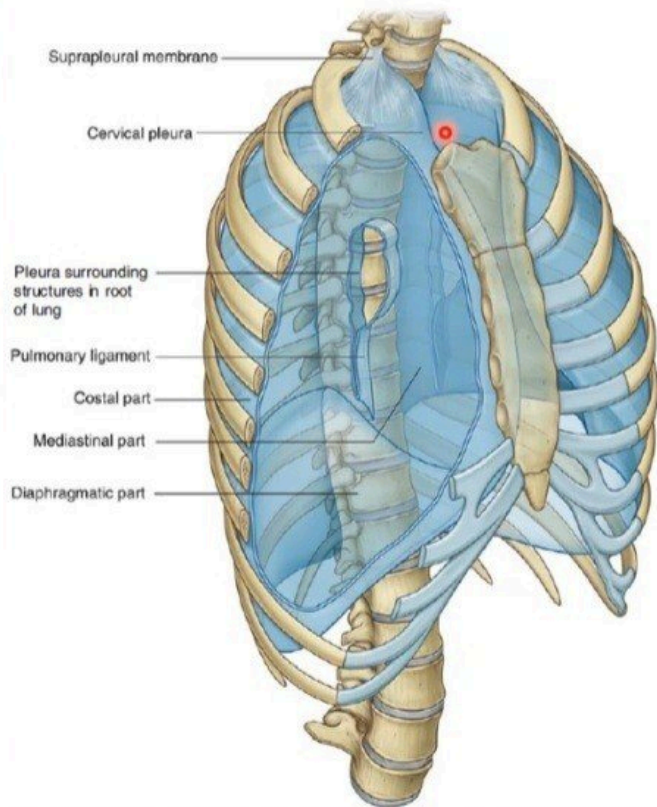
Collateral branches of
intercostal nerve and vessels

Anterior intercostal artery and vein

Anterior perforating
branches of
intercostal vessels







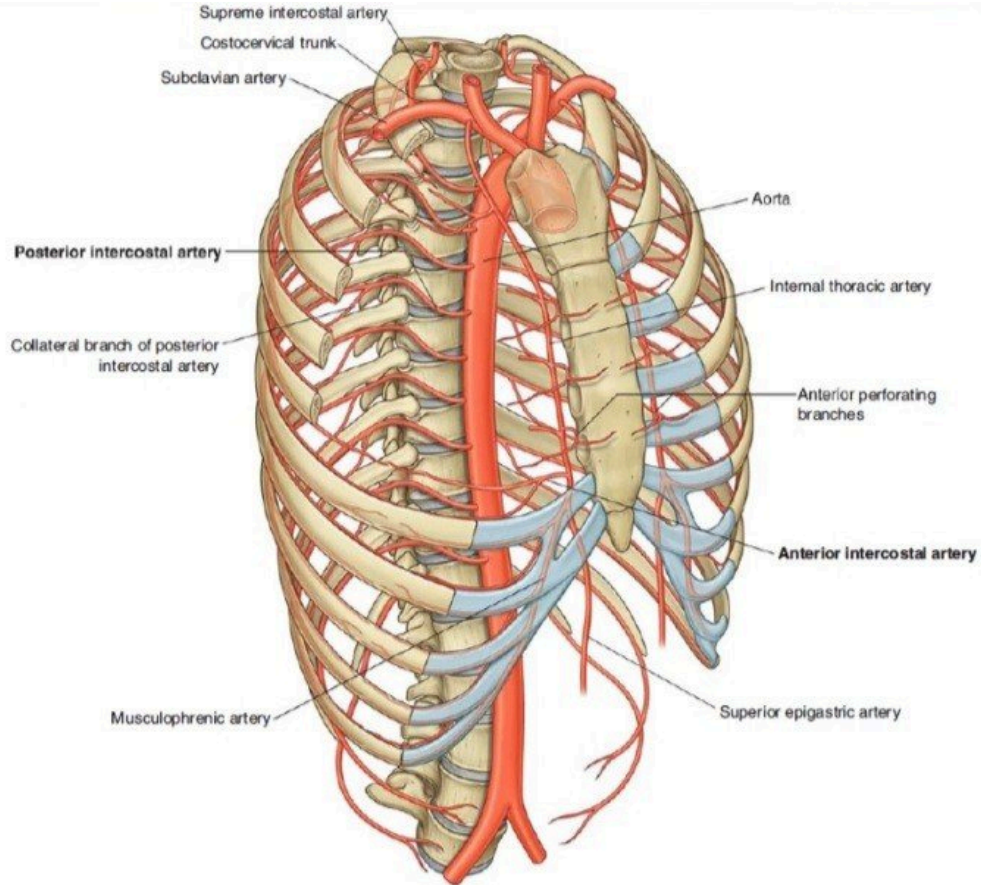
The arterial supply to the thoracic wall

The posterior intercostal arteries of:

- ✓ The **first two** spaces are branches from the **superior intercostal artery**, a branch of the **costocervical trunk** of the subclavian artery.

The posterior intercostal arteries of:

- ✓ the **lower nine** spaces are branches of the **descending thoracic aorta**.



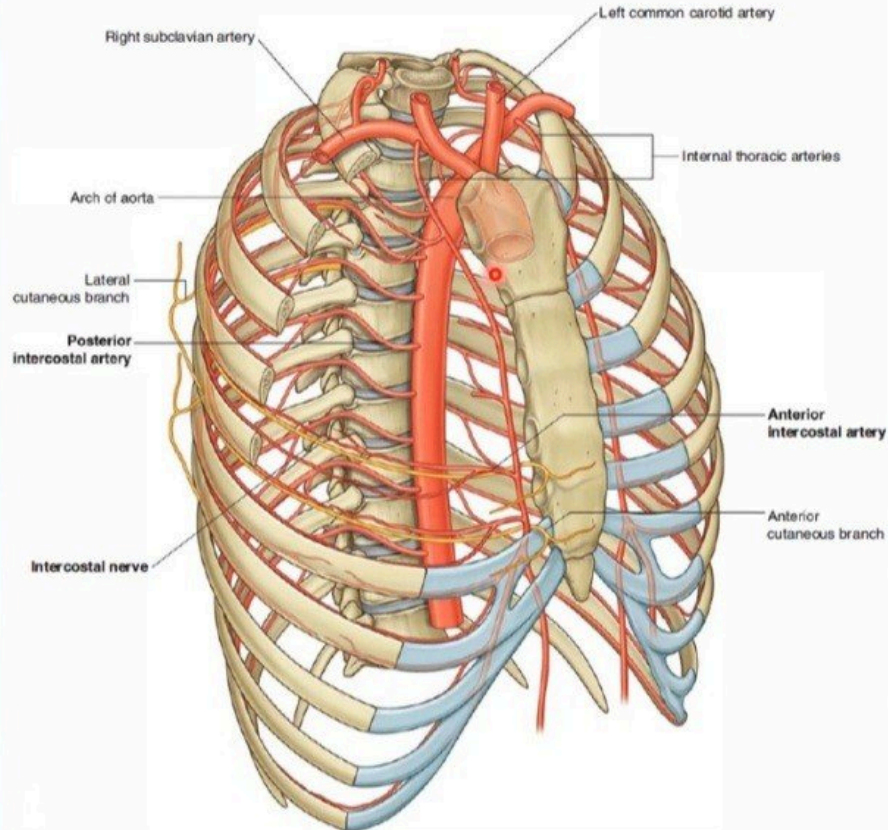
The arterial supply to the thoracic wall

The anterior intercostal arteries of:

- ✓ the **first six spaces** are branches of the **internal thoracic artery**.
 - which arises from the first part of the subclavian artery.

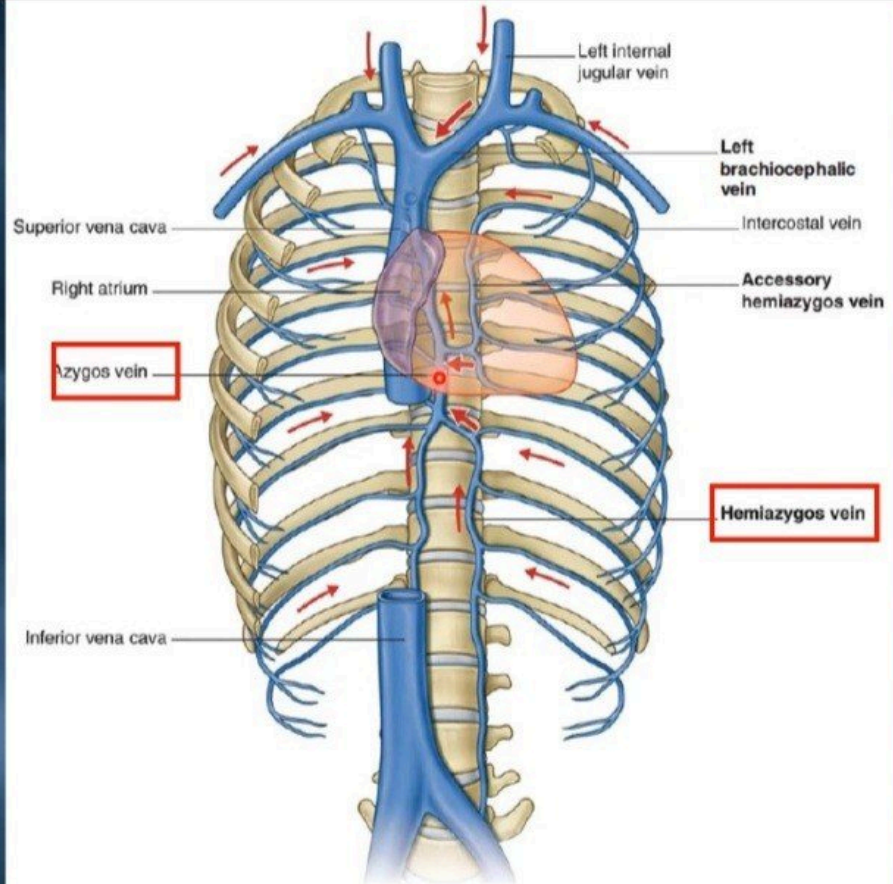
The anterior intercostal arteries of

- ✓ the **lower spaces** are branches of the **musculophrenic artery**,
 - one of the terminal branches of the internal thoracic artery.



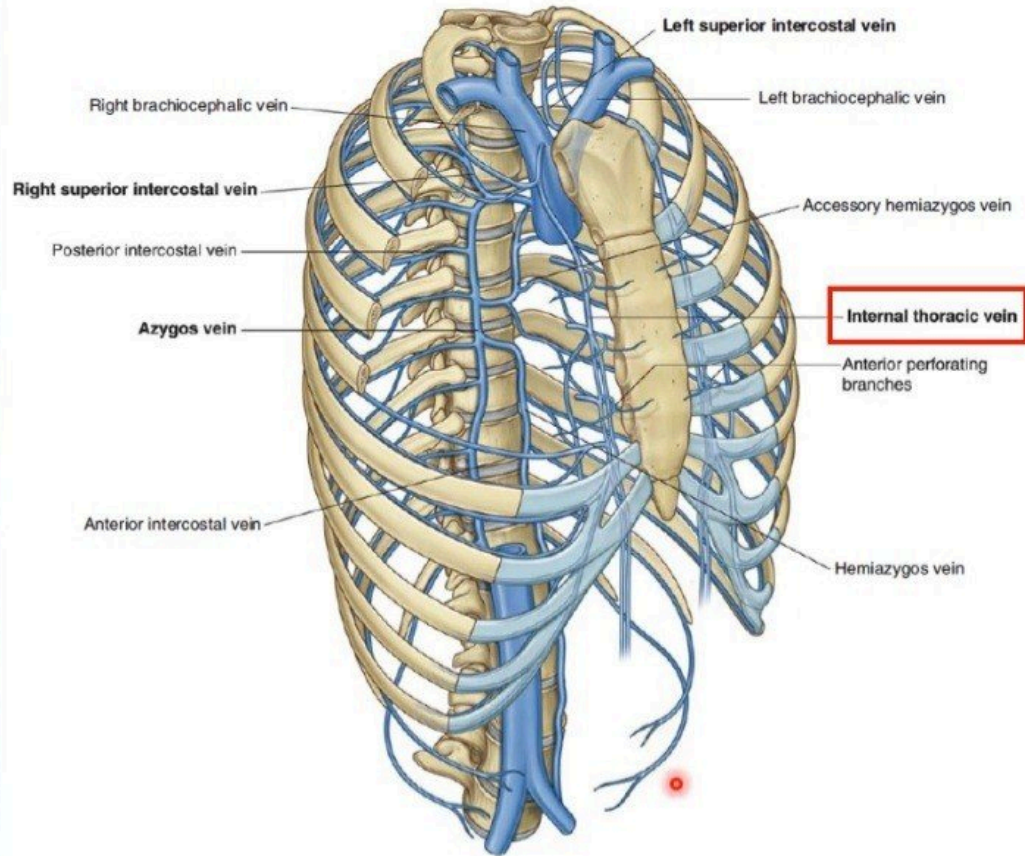
The venous drainage of the thoracic wall

The corresponding posterior intercostal veins drain backward into **the azygos or hemiazygos veins**



The venous drainage of the thoracic wall

and the anterior intercostal veins drain forward into **the internal thoracic** and the musculophrenic veins.

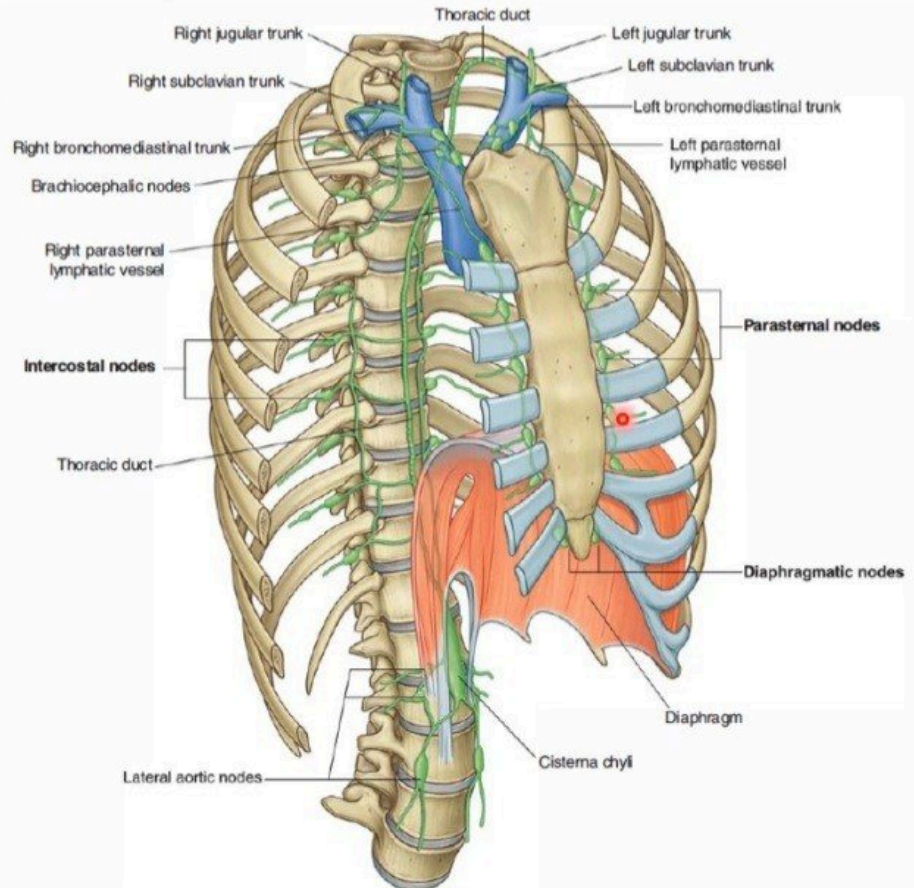


lymphatic drainage of the thorax

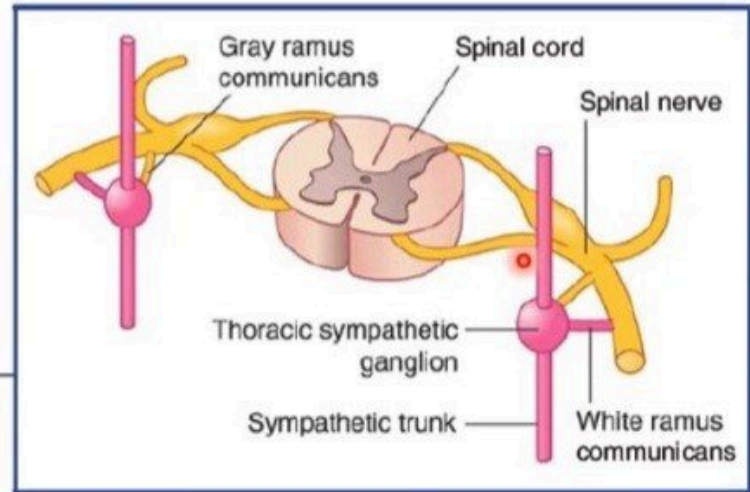
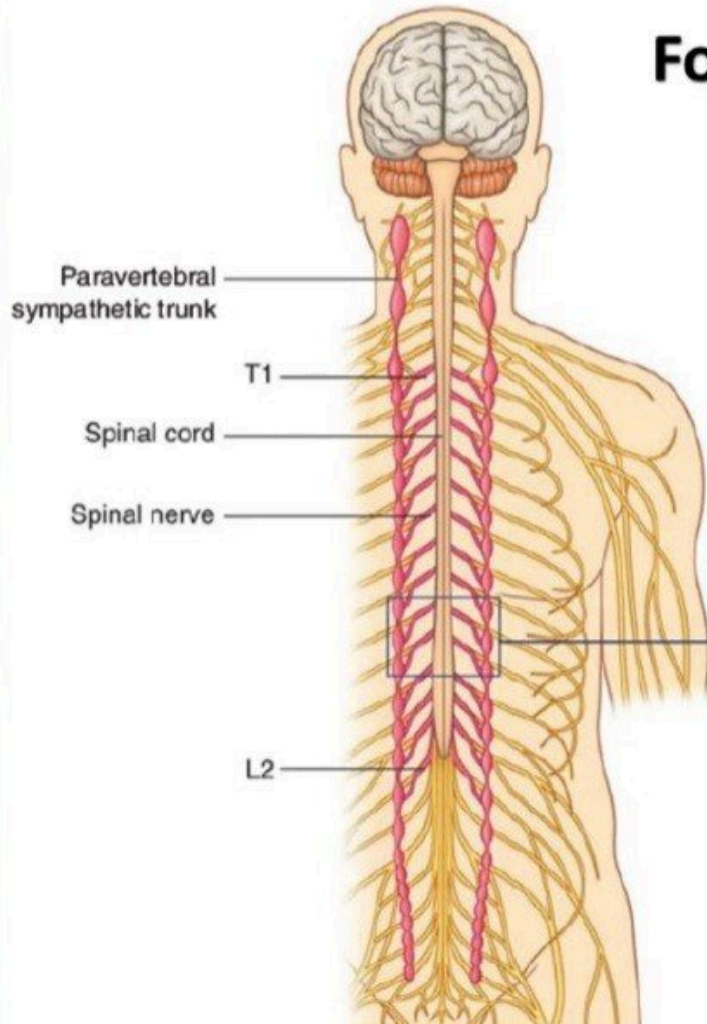
1) Sternal or Parasternal (Internal Thoracic) Nodes

2) Intercostal Nodes

3) Phrenic Nodes



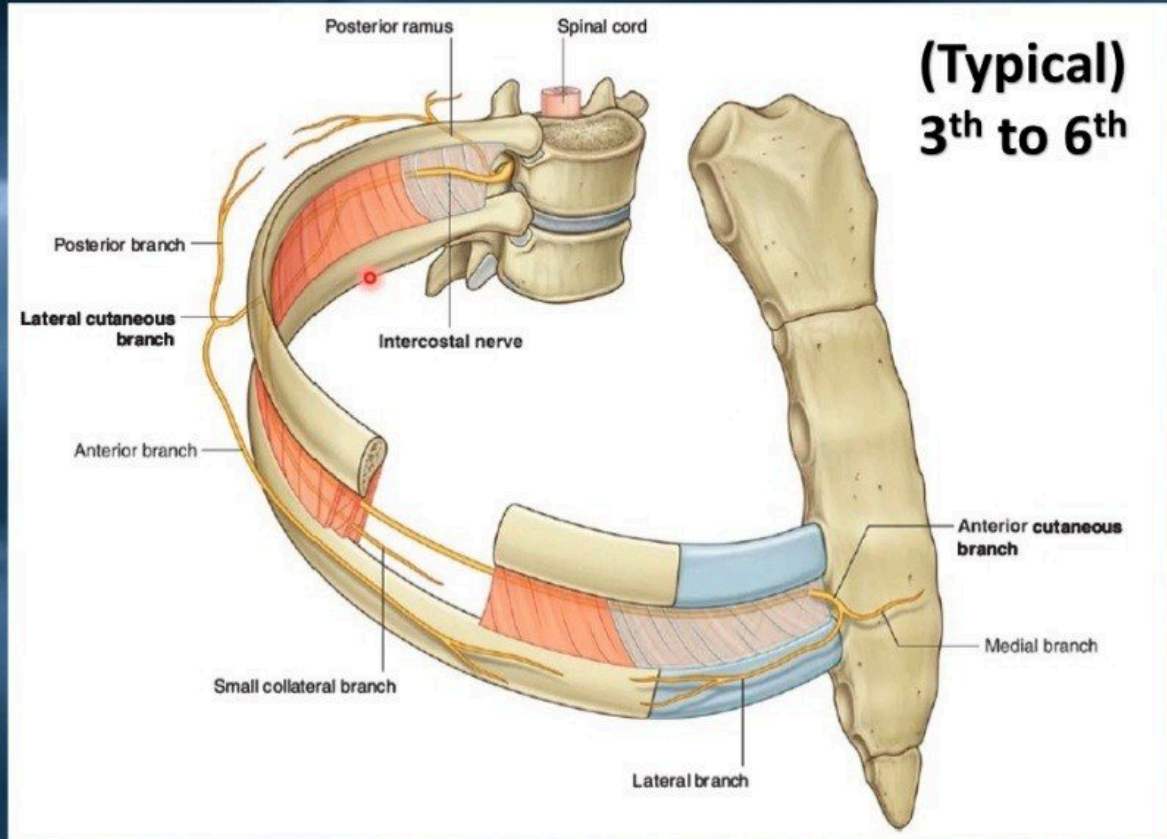
Formation of Intercostal nerve



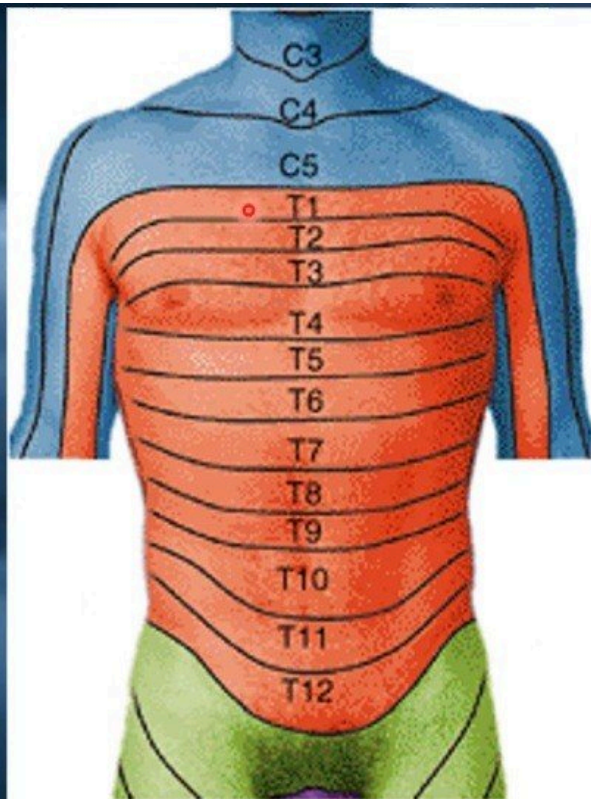
Intercostal nerve

Atypical Intercostal Nerves :

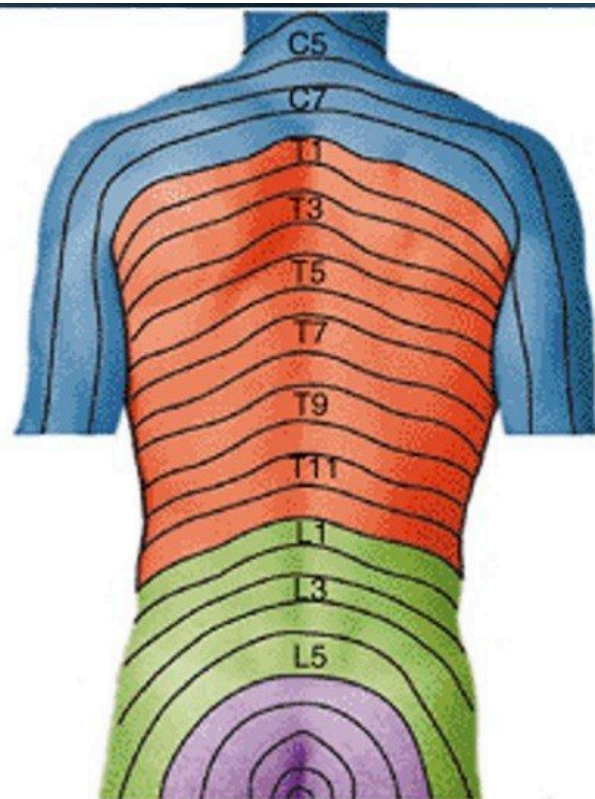
- ✓ First, Second intercostal nerves, To become intercostobrachial nerve.
- ✓ 7th 11th intercostal nerves supply abdominal skin and muscles, To become the thoracoabdominal nerves.



Segmental
innervation
(dermatomes)
of thoracic
wall



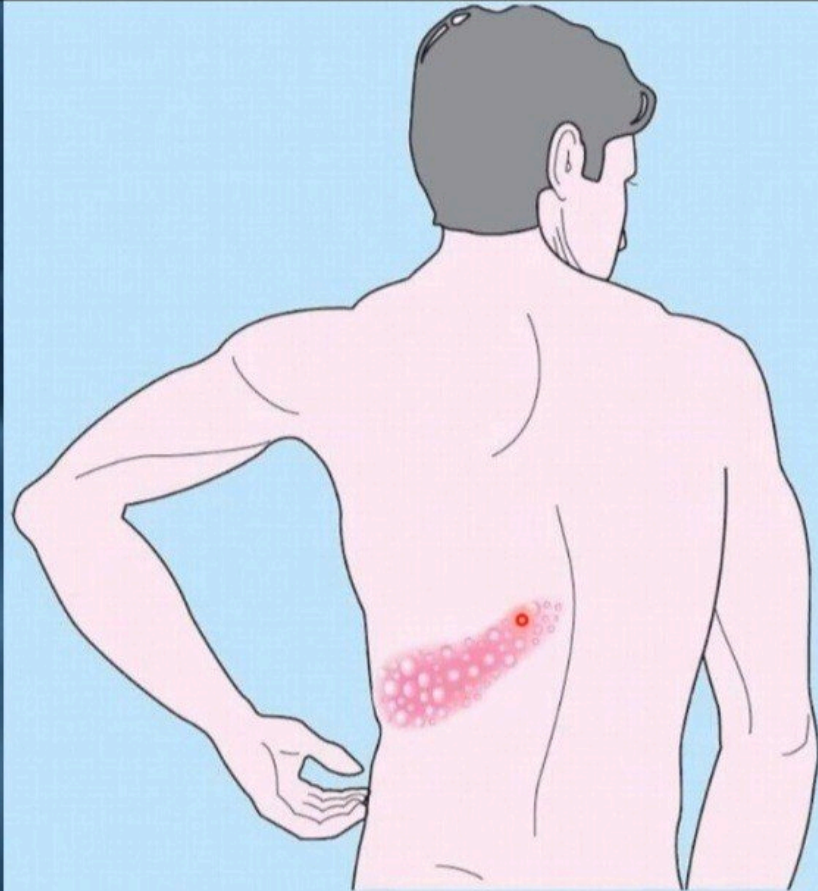
Anterior view



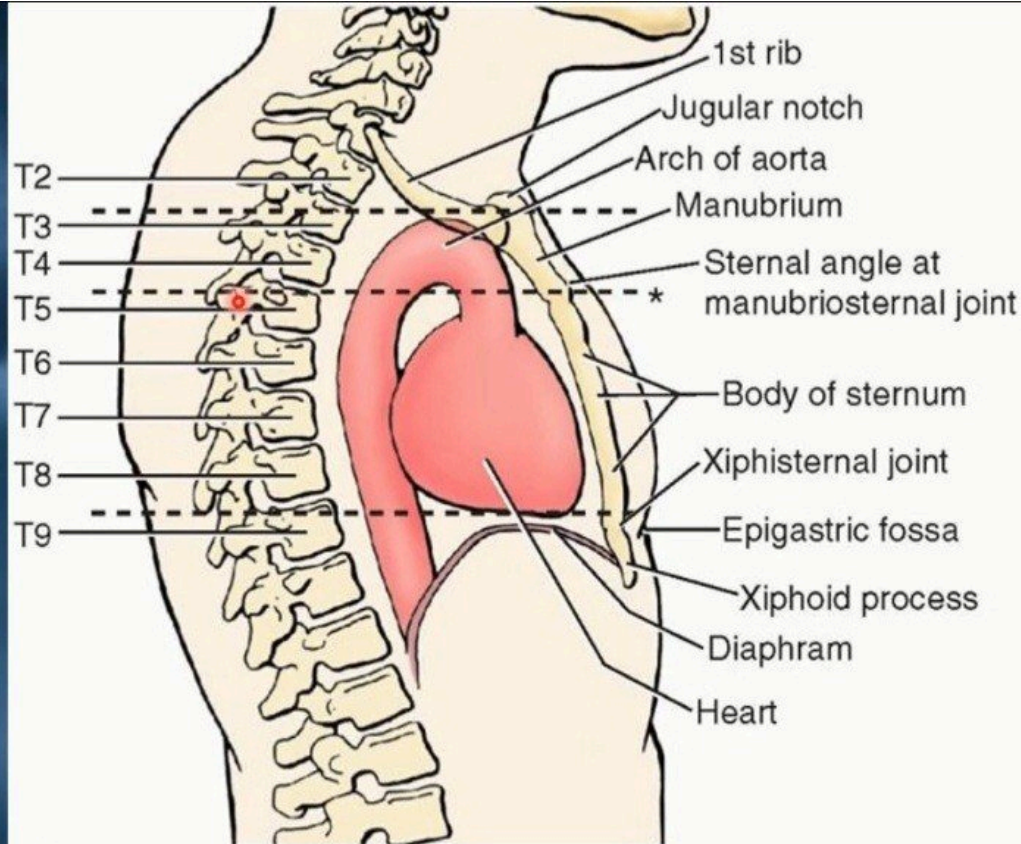
Posterior view

herpes zoster the Spinal Ganglia

- A herpes zoster infection causes a classic, dermatomally distributed skin lesion "shingles" an agonizingly painful.

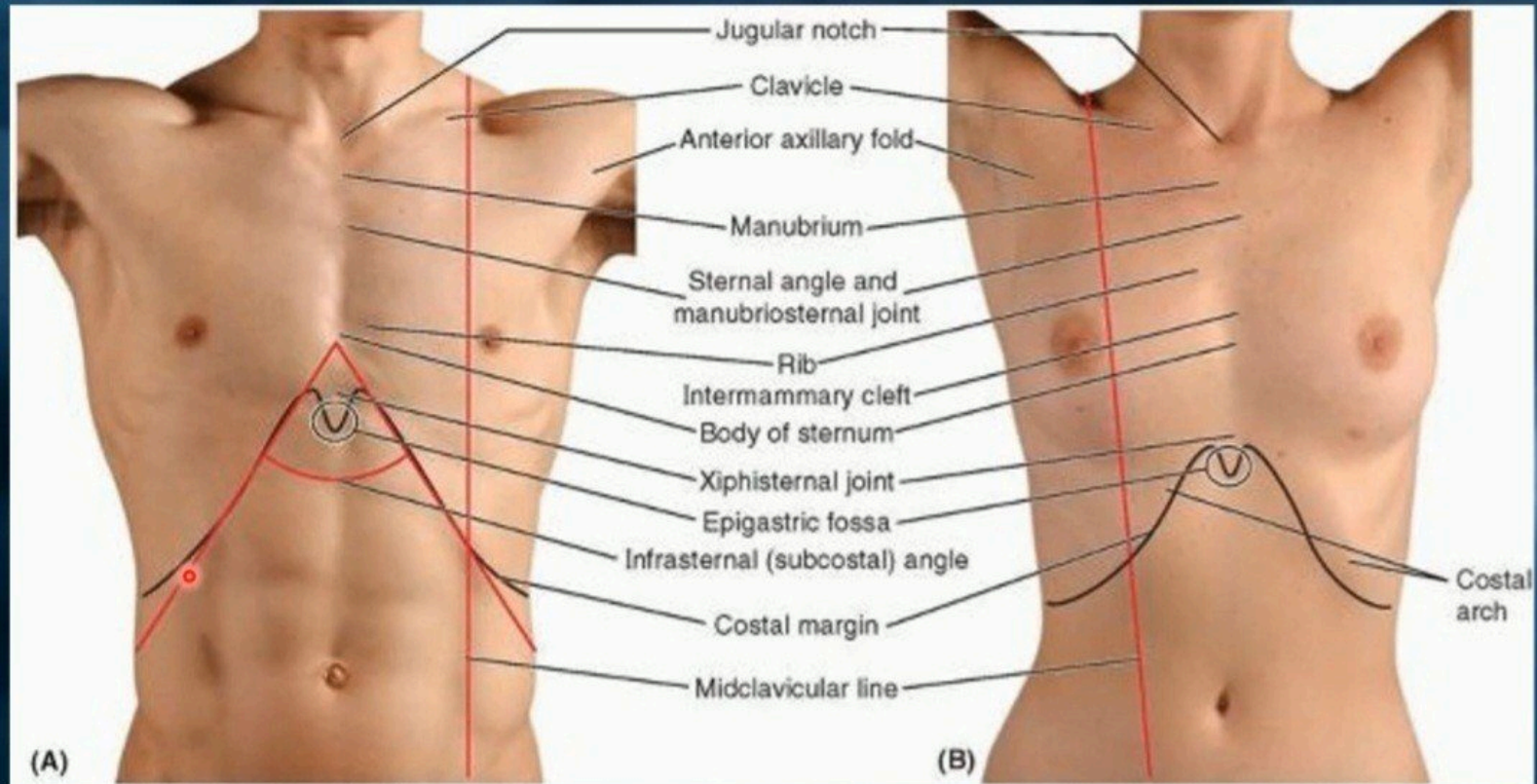


Vertebral levels of sternum and transverse thoracic plane.

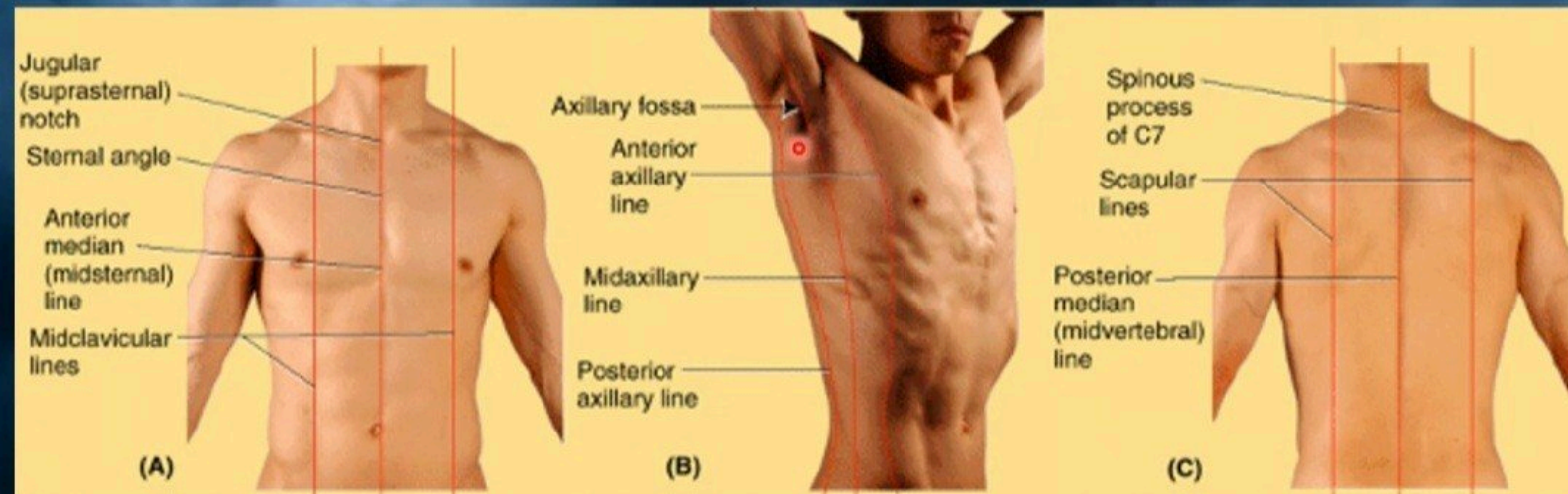


* Transverse thoracic plane

Surface Anatomy of the Thoracic Wall Skeleton



Surface Anatomy of the Thoracic Wall Skeleton





Thanks